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ONE MARK QUESTIONS

# OMQ

## SECTION-A

Be Bold with BBQ!

When Memory becomes Weak, BBQ makes it Peak!

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TARGET MARKS:≥5

## OMQ CHAPTERS

9 MARKS


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☛ Being Bullet Q Book, to make it short only 5 selective OMQ[ 5 MCQ, 5 FIB, 5 OWQ] are taken.

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OMQ – Hints & Solutions

# 1. ELECTRIC CHARGES & FIELDS

## I. MULTIPLE CHOICE Q'S(MCQ)

1. By quantization, charge q of a body is given by (where n is integer) [ 4 ]
   1) $q=n/2e$ 2) $q=n(e/3)$ 3) $q=n(2e/3)$ ✓4) $q=ne$
2. The value of permittivity of free space $\varepsilon_0$ is: [ 3 ]
   1) $9\times10^9\ N\,m^2/C^2$ 2) $8.85\times10^{-12}\frac{Nm^2}{C^2}$ ✓3) $8.85\times10^{-12}\frac{C^2}{Nm^2}$ 4) $9\times10^9\frac{C^2}{Nm^2}$
3. Two charges +2C and +6C are repelling each other with a force of 12N. If each charge is added with -2C, then the value of the force will be [ 4 ]
   1) 4 (Attractive) 2) 4 (Repulsive) 3) 8 (Repulsive) ✓4) Zero
4. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests **[BMP]** [ 3 ]
   1) $E_A > E_B > E_C$ 2) $E_A = E_B = E_C$ ✓3) $E_A = E_C > E_B$ 4) $E_A = E_C < E_B$
5. The angle between the electric dipole moment $\bar{p}$ and the direction of electric field strength due to it on the equatorial line is: [ 3 ]
   1) 0° 2) 90° ✓3) 180° 4) 60°

## II. FILL IN THE BLANKS (FIB)

6. A simple apparatus to detect charge on a body is the **Gold – Leaf electroscope**.
7. A soap bubble is given a negative charge then its radius **increases**.
8. The value of basic unit of charge is **$1.6\times10^{-19}$ C**.
9. The SI unit of electric field is **N/C (or) Volt/m**.
10. SI unit of volume charge density is **$C/m^3$**.

## III. ONE WORD ANSWER Q'S (OWQ)

11. What is the nature of force between two like charges?
    A: The nature of the force between two like charges is **repulsive.**
12. How many number of electrons are in 1 coulomb charge?
    A: Given that $q = 1C$, $e = 1.6\times10^{-19}$ C, $n = ?$
    **Formula:** $q = ne \Rightarrow n = \frac{q}{e} \Rightarrow n = \frac{1}{1.6\times10^{-19}} = 0.625\times10^{19}$ **electrons**
13. What is the direction of electric field due to a single positive charge?
    A: The direction of electric field due to a single positive charge is **radially outward in all directions.**
14. What is the SI unit of electric flux?
    A: SI unit of electric flux is the **Volt-meter (V-m).** It is also equal to $Nm^2/C$.
15. What is the SI unit of electric dipole moment?
    A: SI unit of dipole moment is **coulomb -meter (C-m)**

# 2. CURRENT ELECTRICITY

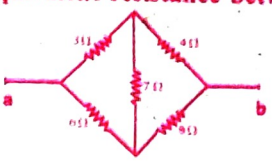
## I. MCQ

1. **Which of the following material has the largest resistivity?** [ 3 ]
   1) Silicon 2) Germanium ✓3) **Glass** 4) Silver

2. **An electric bulb is rated 60 W, 220 V. The resistance of the filament is close to** [ 1 ]
   ✓**1) 807 Ω** 2) 870 Ω 3) 708 Ω 4) 780 Ω

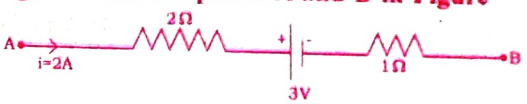
3. **The internal resistance of a 2.1 V cell which gives a current of 0.2 A through the resistance of 10 Ω is** [ 4 ]
   1) 0.8 Ω 2) 1.0 Ω 3) 0.2 Ω ✓**4) 0.5 Ω**

4. **A bridge circuit is shown in figure . The equivalent resistance between points a and b is** [ 4 ]




   1) 21 Ω 2) 7 Ω
   3) $\frac{252}{85}\Omega$ ✓**4)** $\frac{14}{3}\Omega$

5. **The potential difference ($V_A - V_B$) between the points A and B in Figure** [ 4 ]
   1) −3V 2) +3V
   3) +6V ✓**4) +9V**




## II. FIB

6. The electric current in an electrolytic solution is due to the flow of **ions**.
7. The SI unit of resistivity is **Ωm** .
8. The SI unit of current density is **$A/m^2$** .
9. The reciprocal of resistivity is **conductivity**.
10. In an open circuit, the potential difference between the ends of a battery is equal to its **electromotive force (emf)**. **[BMP]**

## III. OWQ

11. **What is the SI unit of electric current?**
    A: SI unit of electric current is the **ampere (A).**

12. **What is the SI unit of Resistance?**
    A: SI unit of electric resistance is the **ohm (Ω).**

13. **What is conductivity?**
    A: **Conductivity** is the ability of a material **to allow the flow of electric charge.**

14. **Give the relation between emf and potential difference in open circuit.**
    A: In an open circuit, **EMF is equal to the potential difference** across the terminals (E = V).

15. **What are the dimensions of Resistivity?**
    A: The dimensional formula of resistivity is $[ML^3T^{-3}A^{-2}]$

# 3. MAGNETISM & MATTER

## I. MCQ

1. The magnetic field lines of the bar magnet looks similar to the magnetic field lines of _____ [ 3 ]

1) a current carrying circular coil 2) a straight long current carrying wire
✓3) **a long solenoid** 4) a plane current sheet

2. The universal character of every material is **[BMP]** [ 1 ]

✓1) **diamagnetic** 2) paramagnetic 3) ferromagnetic 4) ferrimagnetic

3. Which of the following materials exhibits paramagnetic property? [ 3 ]

1) Iron 2) Cobalt ✓3) **Aluminium** 4) Nickel

4. A freely suspended magnet align in _____ direction. [ 3 ]

1) South-West 2) East-West ✓3) **North-South** 4) North-East

5. Which one among the following materials exhibits higher magnetic susceptibility? [ 1 ]

✓1) **Ferromagnetic** 2) Paramagnetic
3) Diamagnetic 4) Both Paramagnetic and Diamagnetic

## II. FIB

6. The SI unit of magnetisation (M) is **ampere/metre**.
7. The SI unit of magnetic field intensity (H) is **ampere/metre**.
8. The magnetic material whose susceptibility is positive but small is **paramagnetic**.
9. Domains are the characteristic feature of **ferro** magnetic materials
10. The net magnetic flux passing through any closed surface around bar magnet is **zero**.

## III. OWQ

11. What is the magnetic field induction at a distance of r from the center of the bar magnet of magnetic moment m on its bisector line (equatorial line)?

A: Along the equatorial line, for short magnet $\mathbf{B_{eq}} = \frac{\mu_0}{4\pi}\frac{\mathbf{M}}{r^3}$. M = magnetic moment, r = distance

---

12. What is the analogy to permittivity $\epsilon_0$ of free space in magnetic field?

A: **Permeability of free space ($\mu_0$).** [$\mu_0 = 4\pi \times 10^{-7}$ H/m.]

---

13. Write a relation between susceptibility and relative permeability.

A: $\mu_r = 1 + \chi$ is a relation between relative permeability ($\mu_r$) and magnetic susceptibility ($\chi$).

---

14. Which law confirms that no mono magnetic poles exist?

A: **Gauss's law for magnetism** is the law that confirms that no mono magnetic poles exist.

---

15. Write down the relation between magnetic induction (B) and magnetic Intensity (H).

A: $B = \mu H$ (or) $B = \mu_0\mu_r H$
This is the relation between magnetic induction (B) and magnetic field intensity (H).

# 4. ELECTRO MAGENTIC INDUCTION

## I. MCQ

1. **Lenz's law is a consequence of which principle?** [ 1 ]
   ✓**1) Conservation of energy** 2) Newton's first law
   3) Coulomb's law 4) Ohm's law
2. **The SI unit of self inductance is:** [ 4 ]
   1) Tesla 2) Weber 3) Ampere ✓**4) Henry**
3. **Mutual induction between two coils does not depend on:** [ 4 ]
   1) number of turns in both coils 2) cross sectional area of coils
   3) magnetic permeability of medium between them ✓**4) current flowing through the coils**
4. **A change in magnetic flux of 0.3 Wb in 0.1 s produces an average emf of:** [ 2 ]
   1) 0.03 V ✓**2) 3 V** 3) 30 V 4) 300 V
5. **If the flux linkage in a coil changes from 2 Wb to 8 Wb in 0.5 s, the average emf induced is:** **[BMP]** [ 2 ]
   1) 10 V ✓**2) 12 V** 3) 6 V 4) 4 V

## II. FIB

6. According to Faraday's law, the induced emf in a coil is directly proportional to the **rate of change** of the magnetic flux.
7. The SI unit of magnetic flux is **Weber**.
8. The back emf in an inductor is always in the direction opposite to the **original emf**.
9. The flux linkage through a coil of N-turns is proportional to the **current** passing through the coil.
10. The direction of induced current is determined by **Lenz** law.

## III. OWQ

11. **Which law states that the induced current opposes the change in flux that produces it?**
A: **Lenz's law**
It states that the direction of an induced current is always such as to oppose the change that causes it.

---

12. **Give the expression for the emf induced between the ends of a metal conductor moving perpendicular to a uniform magnetic field.**
A: **$E = Blv$.** Here $l$ is the length, v is the velocity, B is the uniform magnetic field

---

13. **What is the working principle of a transformer?**
A: Transformer works on the principle of **mutual inductance.**

---

14. **Why are resistance coils double wounded in opposite direction?**
A: To **minimize/ avoid self inductance.**

---

15. **What is the unit of mutual inductance?**
A: SI unit of mutual inductance is **Henry** (H)

# 5. ALTERNATING CURRENT

## I. MCQ

1. **The frequency of AC mains in India is:** [ 2 ]
   1) 25 Hz ✓**2) 50 Hz** 3) 100 Hz 4) 60 Hz
2. **In a series LCR circuit at resonance, the current is:** [ 2 ]
   1) minimum ✓**2) maximum** 3) zero 4) infinite
3. **The reactance of a capacitor decreases with:** [ 2 ]
   1) decrease in capacitance ✓**2) increase in frequency**
   3) both (1) and (2) 4) decrease in frequency
4. **The power factor of a series LCR circuit at resonance is:** [ 3 ]
   1) 0 2) 0.5 ✓**3) 1** 4) −1
5. **A transformer is designed to:** [ 1 ]
   ✓**1) increase or decrease AC voltage** 2) convert AC into DC
   3) increase the frequency of AC 4) store electric charge

## II. FIB

6. The r.m.s. value of an AC of peak value $I_0$ is $\underline{I_0 / \sqrt{2}}$.
7. The phase difference between current and voltage in a purely resistive AC circuit is $\underline{0°}$.
8. In an LCR circuit at resonance, the impedance is equal to **resistance (R)**.
9. In a transformer, the core is laminated to reduce **eddy current** losses.
10. The power factor of an AC circuit is given by $\underline{\cos\phi}$.

## III. OWQ

**11. What is the r.m.s value of an AC whose peak value is 141 V?**

A: $V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141}{1.414} = 99.7V = \mathbf{100V}$ [∵peak value $V_0 = 141V$]

---

**12. What is the time period of AC mains in India?** **[BMP]**

A: The time period (T) of AC mains in India is **0.02 seconds.**

∴Time period $T = \frac{1}{\text{frequency}} = \frac{1}{50} = 0.02s = 20$ milli seconds [∵standard AC frequency in India is 50Hz.]

---

**13. What is the condition for resonance in a series LCR circuit?**

A: Condition for resonance: $X_L = X_C$ i.e., Inductive reactance = Capacitive reactance.

---

**14. Which device is used to reduce AC without much loss of energy?**

A: **Step down transformer** is the device used to reduce AC without much loss of energy.

---

**15. Write the expression for power factor in an AC circuit.**

A: Power factor $\cos\phi = \frac{R}{Z}$ ; R= Resistance, Z= Impedance, $\phi$ is phase angle.

# 6. ELECTRO MAGNETIC WAVES

## I. MCQ

1. **Which colour of the light has the longest wavelength?** [ 3 ]
   1) Green 2) Blue ✓**3) Red** 4) Violet
2. **If the frequency of an electromagnetic wave is 60 MHz and it travels in free space, the wavelength of the wave (m) is:** [ 2 ]
   1) 2.5 ✓**2) 5** 3) 10 4) 2
3. **Green House effect is due to** [ 4 ]
   1) absorption of Ultra-Violet radiation by Ozone layer
   2) transmission of visible light by the atmosphere
   3) good weather
   ✓**4) reflection of infrared rays emitted by earth's atmosphere**
4. **Ozone Layer absorbs** [ 2 ]
   1) Visible light ✓**2) UV radiation** 3) Infrared radiation 4) Microwaves
5. **Which of the following waves are also called the "Heat waves"?** [ 3 ]
   1) Visible light 2) X-rays ✓**3) Infrared rays** 4) Radio waves

## II.FIB

6. The spectrum of electromagnetic waves stretching in order of increasing wavelength from **Gamma** rays to long **Radio** waves.
7. In the charging capacitor both conduction current and **displacement** current may be present.
8. Faraday's law of induction states that there is an induced emf equal to the rate of change of **magnetic flux**.
9. **Accelerated** charges radiate electromagnetic waves. **[BMP]**
10. Jagadish Chandra Bose succeeded in producing electromagnetic waves of **shorter** wavelength.

## III. OWQ

11. **In cellular phones which band of radio waves are used to transmit voice communication?**
A: Cellular phones use UHF band **(Ultra High Frequency)** .
The frequency range is 300 MHz to 3 GHz.

---

12. **Earth surface absorbs the incoming visible radiation from the sun and what type of waves it re-radiate?**
A: Earth re-radiates **long infrared waves.**

---

13. **What type of rays can snakes detect in addition to visible rays?**
A: Snakes can detect **infrared rays in** addition to visible rays.

---

14. **When high energy electrons bombard a metal target, what type of waves are produced?**
A: **X-rays** are produced when high energy electrons bombard a metal target.

---

15. **Which type of electromagnetic waves are used in the treatment of cancer?**
A: **Gamma rays** and **X-rays** are used in radiation therapy to destroy cancer cells.

# 7. WAVE OPTICS

## I. MCQ

1. When three lights of red, violet and green coloured are used successively in young double slit experiment, the fringe width will be maximum for ..... [ 1 ]

✓1) Red 2) Violet 3) Green 4) Same for all

2. The phase difference between two points 'A' and 'B' on a wave front separated by distance λ is [ 2 ]

1) π/4 ✓2) 2π 3) π 4) π/2

3. The penetration of light into the region of geometrical shadow is called [ 3 ]

1) polarization 2) interference ✓3) diffraction 4) Refraction

4. Which one of the following phenomenon confirms that transverse nature of light? [ 4 ]

1) Interference 2) Diffraction 3) Dispersion ✓4) Polarization

5. Which of the following waves cannot be polarized? [ 3 ]

1) X- rays 2) Radio waves ✓3) Sound waves 4) Light waves

## II. FIB

6. A wave front is the continuous locus of all particles vibrating with **same phase**.
7. The secondary wave fronts emanating from a primary wave front are also called as **wavelets**.
8. In interference pattern, all bright and dark fringes are of **same** width.
9. For constructive interference, the path difference between two waves is equal to **nλ** .
10. Due to **diffraction** phenomenon of light, colours on CD are formed.

## III.OWQ

11. State Huygen's principle.

A: Every point on a wave front acts as a source of secondary spherical wavelets.

---

12. Which principle is used to find the position of a wave front after some time? **[BMP]**

A: **Huygen's principle** is used to find the position of a wave front.

---

13. What is interference of light?

A: Interference is the phenomenon where two or more light waves overlap to form a resultant wave of greater, lower or the same amplitude.

---

14. What is the main condition to produce interference of light?

A: The primary condition is that the **light sources must be coherent.**

---

15. What do you mean by the polarisation of light?

A: Polarization is the process of restricting the vibrations of light to a single plane perpendicular to the direction of propagation.

# 8. ATOMS

## I. MCQ

1. **1eV is equivalent to** [ 3 ]
   1) $6.67\times10^{-34}$ J 2) $1.6\times10^{-27}$ J ✓3) $1.6\times10^{-19}$ J 4) $6.24\times10^{-22}$J

2. **The total energy of an electron in the stationary states of the hydrogen atom is** [ 2 ]
   1) $E_n = -\frac{13.6}{n}$eV ✓2) $E_n = -\frac{13.6}{n^2}$eV 3) $E_n = \frac{13.6}{n}$eV 4) $E_n = \frac{13.6}{n^2}$eV

3. **Bohr's atomic model is not valid for** [ 4 ]
   1) hydrogen atom 2) singly ionized helium atom ($He^+$)
   3) deuterium atom ✓4) **singly ionized neon atom ($Ne^+$)**

4. **The angular momentum of an electron revolving in $n^{th}$ orbit is given by** [ 1 ]
   ✓1) **$mvr = nh/2\pi$** 2) $mvr = n^2h/2\pi$ 3) $mvr = h\pi n^3$ 4) $mvr = n^2h^2/2\pi$

5. **According to Bohr atomic model, in which of the following transition will the frequency be maximum?** [ 2 ]
   1) n = 4 to n = 3 ✓2) **n = 2 to n = 1** 3) n = 5 to n = 4 4) n = 3 to n = 2

## II. FIB

6. A $He^+$ ion is in its first excited state, its ionization energy is **13.6 eV**
7. The change in angular momentum of an electron when it jumps from $4^{th}$ orbit to $5^{th}$ orbit in hydrogen atom is **$h/2\pi$**.
8. The process of raising or transferring the electron from lower energy state to higher energy state is called **excitation**. **[BMP]**
9. The process of raising the atom from the normal state to the ionized state is called **ionization**.
10. 1 fermi is equal to **$10^{-15}$** meter.

## III. OWQ

11. **What kind of charge is spread throughout the atom in Thomson's model?**
    A: In Thomson's model, **positive charge** is spread throughout the atom.

---

12. **What is the SI unit of the impact parameter?**
    A: The SI unit of impact parameter is **metre (m).**

---

13. **The energy of a hydrogen atom in the ground state is –13.6eV. What will be the energy of $He^+$ ion in the first excited state?**
    A: For $He^+$ ion we have Z=2 and for the first excited state n=2
    Energy $E_n = \frac{-13.6Z^2}{n^2} \Rightarrow E_2 = \frac{-13.6(2)^2}{(2)^2} = -13.6$eV.

---

14. **An electron in a hydrogen atom revolving in its second excited state having the radius of 4.65Å. What is its de-Broglie wavelength?**
    A: According to Bohr's postulate, $2\pi r = n\lambda \Rightarrow \lambda = \frac{2\pi r}{n} = \frac{2\times3.14\times4.65}{3} = 9.734$Å

---

15. **What is the path of an electron around the nucleus as per Rutherford's atomic model?**
    A: According to Rutherford's atomic model, the path of an electron around the nucleus is a **circular orbit.**

# 9. SEMI CONDUCTORS

## I. MCQ

1. **Which of the following is used as pentavalent impurities for doping process?** [ 1 ]
   ✓**1) Arsenic** 2) Boron 3) Indium 4) Aluminium

2. **The energy band gaps of silicon and germanium are** [ 3 ]
   1) 0.7eV, 1.1eV 2) 1eV, 0.7eV ✓**3) 1.1eV,0.7eV** 4) 2eV, 1.5eV

3. **In case of full wave rectification, if the input frequency is 60 Hz, then the output frequency would be** [ 2 ]
   1) 60 Hz ✓**2) 120 Hz** 3) 100 Hz 4) 150 Hz

4. **For a p-n junction diode, the current in reverse bias is in the order of** [ 3 ]
   1) Few mA 2) Few amperes ✓**3) Few μA** 4) Few kilo amperes

5. **The increase in the width of the depletion region in a p-n junction diode is due to** [ 1 ]
   ✓**1) reverse bias only** 2) both forward bias and reverse bias
   3) high doping concentrations 4) forward bias only

## II. FIB

6. In a vacuum tube, the electrons are supplied by a **cathode**.
7. In vacuum tubes, electrons flow from the cathode to the anode in **only one** direction.
8. The gap between the top of the valence band and bottom of the conduction band is called the **energy band gap/forbidden energy gap**. **[BMP]**
9. The pentavalent dopant donates one extra electron for conduction and hence this impurity is called **donor**.
10. In the p-n junction diode symbol, the direction of arrow indicates **direction of conventional current**.

## III. OWQ

11. **At what temperature the intrinsic semiconductor will behave like an insulator?**

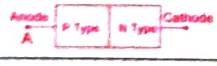
A: **At absolute zero temperature (0 K)** intrinsic semi conductor behaves like an insulator

---

12. **Draw the circuit symbol of the p-n junction diode.**

A: **Circuit symbol of the p-n junction diode:**

**p-n junction diode:**

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13. **What is the forbidden energy gap in Ge semiconductor crystal in terms of Joules?**

A: **Forbidden energy gap in Ge** at room temperature $Eg=0.72\times1.6\times10^{-19}$ J $=\mathbf{1.15\times10^{-19}}$ **J**

---

14. **What is the name of the current due to motion of charge carriers when external electric field is applied.**

A: The current produced by the motion of charge carries under the influence of an external electric field is known as **drift current.**

---

15. **What is the name of the process "adding impurities to the intrinsic semiconductor to make extrinsic semiconductor"?**

A: **Doping** is the process of adding impurities to an intrinsic semiconductor to make extrinsic semi conductor.

SHINING STARS
# STAR QUESTIONS PLUS

## ELECTRIC CHARGES & FIELDS
## VSAQ

**182. What happens to the force between two charges if the distance between them is a) halved b) doubled**

A : From Coulomb's law, we have $F \propto \frac{1}{r^2} \Rightarrow \frac{F_2}{F_1} = \frac{r_1^2}{r_2^2}$

a) When distance is halved $r_2 = \frac{r_1}{2}$; New force $F_2 = ?$

$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} = \frac{r_1^2}{(r_1/2)^2} = \frac{r_1^2}{r_1^2/4} = 4 \Rightarrow F_2 = 4F_1$. Thus, force **increases to 4 times.**

b) When distance is doubled $r_2 = 2r_1$, $F_2 = ?$

$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} = \frac{r_1^2}{(2r_1)^2} = \frac{r_1^2}{4r_1^2} = \frac{1}{4} \Rightarrow F_2 = F_1/4$. Thus, the force **decreases to 1/4.**

---

**183. When will be the electric flux negative and when is it positive?**

A : **Electric flux is negative** due to a **negative charge** and **positive** due to a **positive charge.**

**Reason:** Electric flux $\Delta\phi = \vec{E} \cdot \Delta\vec{S} = E\Delta S\cos\theta$

Electric flux $\Delta\phi$ is negative when cos θ is negative i.e,, when θ belongs to $Q_2$ or $Q_3$.

Electric flux $\Delta\phi$ is positive when cos θ is positive i.e,, when θ belongs to $Q_1$ or $Q_4$.

---

**184. Write the expression for electric intensity due to a charged conducting spherical shell at points outside and inside the shell. Explain the terms in it.**

A : 1) Electric intensity outside a charged conducting spherical shell is $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

Here q is the total charge on the spherical shell, r is the distance of the point outside the shell and $\varepsilon_0$ is permittivity of free space.

2) Electric intensity inside the shell is zero. i.e., E = 0

---

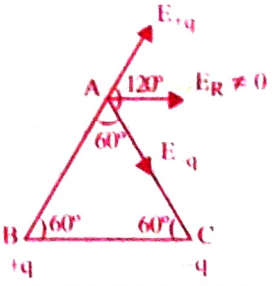
**185. Consider two charges +q and -q placed at B and C of an equilateral triangle ABC. For this system, the total charge is zero. But the electric field (intensity) at A which is equidistant from B and C is not zero. Why ?**

A : At point A, the electric field vectors due to +q and −q are not in opposite direction. (actually 120° ).

So they do not cancel each other.

Hence the resultant vector is non-zero.

So, intensity of electric field is not zero at A.



## ELECTROSTATIC POTENTIAL

**186. What will be the shapes of equipotential surfaces in case of a uniform electric field and a point charge?**

**Sol:** In a uniform electric field, the equipotential surfaces take the shape of **equidistant parallel planes** perpendicular to direction of the electric field lines.

At a point charge, equipotential surfaces are **concentric spheres** centered on the point charge.

---

**187. The dielectric strength of air is $3\times10^6 Vm^{-1}$ at certain pressure. A parallel plate capacitor with air in between the plates has a plate separated by 1 cm. Can you charge the capacitor to $3\times10^6$ V . Give reason.**

**A :** No.

Since air is the dielectric, the **maximum possible** potential of the capacitor is V=Ed

$$V= 3\times10^6 \times10^{-2}=3\times10^4 \text{ V}$$

But required potential is $3\times10^6$ V. Required potential > Possible potential .

So it is not possible for the capacitor to charge potential $3\times10^6$ V .

---

## CURRENT ELECTRICITY

**188. If a wire is stretched to double its original length without the loss of mass, how will the resistivity of the wire be influenced?**

**A :** **1) Resistivity** is the property of material of a wire and it does not change with the change in dimensions (length) of the wire.

**2)** When a wire is stretched to double its length, its resistivity does not change, but its **resistance becomes 4 times**. Because for a given mass, volume is constant . Here $R \propto l^2$

---

**189. The electron drift speed in metals is small (~mm/s) and the charge of the electron is also very small ($\sim10^{-19}$C ), but we can still obtain a large amount of current in a metal. Why?**

**A :** Current in a conductor is $I = nAev_d$

Though the values of e and $v_d$ are very small, the value of number of free electrons per unit volume n is very very large($\sim10^{28}$) for metals. Hence large currents are possible in metals.

# MOVING CHARGES & MAGNETISM

**190. Express a relation among the quantities the speed of light in vacuum, the permeability of vacuum and the permittivity of vacuum.**

A : The speed of light in vacuum $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Here $\mu_0$ is permeability and $\varepsilon_0$ is the permittivity of vaccum.

---

**191. What is the magnitude of magnetic force on a charged particle of strength q moving through a uniform magnetic field B at a velocity v which makes an angle θ with magnetic field? When does this force become maximum?**

A : 1) The magnitude of magnetic force (F) on a charge (q), moving with velocity (v) at an angle θ to a magnetic field (B) is given by **F = Bqv sinθ.**

2) This force becomes maximum when θ=90°. Then sinθ = sin 90° = 1. ∴ $F_{max}$ = Bqv

---

**192. What is the magnitude of magnetic force on a current (i) carrying wire of length 'l' placed in a uniform magnetic field B if it makes an angle θ with magnetic field? When does this force become maximum?**

A : 1) The magnitude of magnetic force (F) on a current (i) carrying wire of length (*l*) placed in a magnetic field (B) which makes an angle (θ), is given by **F = Bi*l* sinθ**

2) This force becomes maximum when θ=90°. Then sinθ = sin 90° = 1. ∴ $F_{max}$ =Bi*l*

---

**193. What is the work done by the magnetic field on moving charge? Give reason.**

A: 1) The work done by a magnetic field on a moving charge is **zero.**

2) **Reason:** Magnetic field $\vec{F}$ is always perpendicular to the velocity $\vec{v}$ of the charge.

Work done $W = \int \vec{F}.\vec{ds} = \int \vec{F}.\vec{v}\, dt$ (dot product becomes zero)

---

**194. Write expression for the magnetic field induction at a point on the axis of a circular current carrying coil. Obtain an expression for the magnetic field induction at its center.**

A 1) The magnetic induction on the axis of a circular coil at a distance x from its centre is

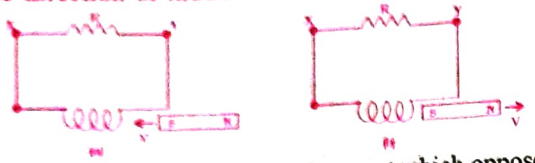
$B_x = \frac{\mu_0 NIR^2}{2(x^2 + R^2)^{3/2}}$ Here N is number of the turns, I is the current passing through the coil, R is radius of the coil, $\mu_0$ is permeability of free space.

**2) Magnetic Induction at the centre of the circular coil:**

Putting x = 0 in the above equation, we get magnetic induction the at centre of the coil $B_0 = \frac{\mu_0 NI}{2R}$

## ELECTRO MAGNETIC INDUCTION

**195. (a) & (b). State the direction of induced current in the resistor 'R'.**



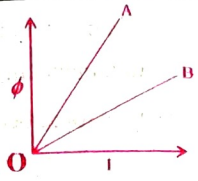
A: According to Lenz's law, the direction of induced current which opposes the cause of it.

a) If the magnet moves **towards** the coil, **flux increases.**
Hence the induced current in the resistor R moves in anti-clockwise direction from Y to X.

b) If the magnet moves **away** from the coil, **flux decreases.**
Hence the induced current in the resistor R moves in clockwise direction from X to Y.

---

**196. A plot of magnetic flux ($\phi$) versus current (I) is shown in figure for two inductors A and B. Which of the two has larger value of self induction. State the reason.**




A: 1) In the given graph, slope of the line represents self inductance (L).

$$\text{Slope} = \frac{\phi}{I} = L$$

2) In the graph, slope of line OA is greater than the slope of the line OB.

Hence, the self inductance of A is greater than the self inductance of B. Thus $L_A > L_B$

## ELECTRO MAGNETIC WAVES

**197. If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon?**

A: Energy of photon $E = \frac{hc}{\lambda} \Rightarrow E\alpha\frac{1}{\lambda}$ Let $\lambda_2 = 2\lambda_1$

$\therefore \frac{E_2}{E_1} = \frac{\lambda_1}{\lambda_2} = \frac{\lambda_1}{2\lambda_1} = \frac{1}{2} \Rightarrow E_2 = \frac{E_1}{2}$. Thus, energy of **photon becomes half.**

---

**198. What is the ratio of speed of infrared rays and ultraviolet rays in vacuum?**

A: Ratio of speed of infrared rays and ultraviolet rays in vacuum is **1:1.**
Because they both travel with same speed of light in vacuum

---

**199. What is the relation between the amplitudes of the electric and magnetic fields in free space for an electromagnetic wave?**

A: Velocity of light $c = \frac{E_0}{B_0}$

Here $E_0$ is amplitude of electric field, $B_0$ is amplitude of magnetic field in free space.

---

**200. The energy of the EM waves is of the order of 15 KeV. To which part of the spectrum does it belong?**

A: The range of x-ray part in EM waves is 0.1 keV to 100 keV. So,15 keV belongs to **x-rays.**

**201. A plane electromagnetic wave of frequency 20 MHz travels in free space along the + X direction. At a particular point in space and time, the electric field vector of the wave is $E_y$ = 9.3 V/m. Then find the magnetic field vector of the wave at that point?**

**A:** Given Electric field $E_y = 9.3 V/m$ in the direction of y-axis.

Velocity of light $c = 3 \times 10^8$ m/s. Waves travel along +ve x-axis

Magnetic field $\vec{B} = \frac{E_y}{c} = \frac{9.3}{3\times10^8} = 3.1\times10^{-8} T = 3.1\times10^{-8}\hat{k}T$

$\hat{k}$ refers to positive z-axis as $\vec{E}, \vec{B}$ and propagation of EM waves are mutually perpendicular.

---

**202. A beam of light travelling along X-axis is described by the electric field $E_y = 900 \sin\omega\left(t - \frac{x}{c}\right)$, then find the ratio of electric force to magnetic force on a charge moving along Y-axis with a speed of $3 \times 10^8$ m/s. (Given speed of light is $3 \times 10^8$ m/s).**

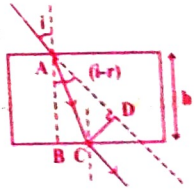
**A:** Force due to electric field $F_E = E_0 q$ ; Force due to magnetic field $F_B = B_0 q v$

$$\therefore \frac{F_E}{F_B} = \frac{E_0 \cancel{q}}{B_0 \cancel{q} v} = \frac{E_0}{B_0 v} = \frac{c\cancel{B_0}}{\cancel{B_0} v} = \frac{c}{v} = \frac{3\times10^8}{3\times10^8} = \frac{1}{1} = 1:1 \qquad (\because E_0 = cB_0)$$

## RAY OPTICS

**203. Discuss the lateral shift of a ray refracted through a parallel side glass slab with neat ray diagram.**

**A:** **Lateral shift** is the perpendicular distance between the incident ray's original path and the emergent ray after passing through a parallel sided glass slab. In the diagram the **lateral shift is CD.**



**Lateral shift of a ray**

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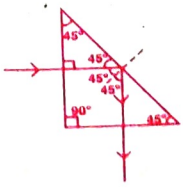
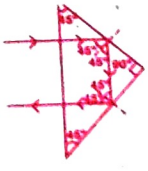
**204. Draw the ray diagrams of crown glass prism to bend rays by 90° and 180° to form an inverted image with the use of total internal reflection.**

**A:**

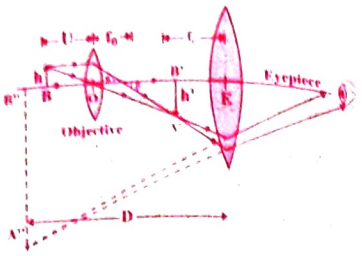


**205. Draw a labelled ray diagram to show the formation of an image by a compound microscope. Write the expression for its magnifying power.**

A: **Formula:** At **near point**, magnifying power

$$m = m_0 m_e = \left[\frac{v_0}{u_0}\right]\left[1 + \frac{D}{f_e}\right]$$




At **infinity**, magnifying power $m = m_0 m_e = \left[\frac{v_0}{u_0}\right]\left[\frac{D}{f_e}\right]$

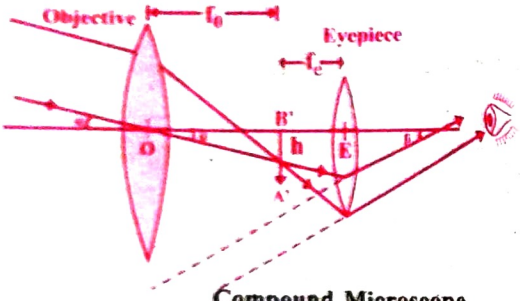
$m_0$ = Magnification of the object; $m_e$ = Magnification of the eye piece, $f_e$ = Focal length of eye piece

D = least distance of distinct of vision; $u_0$ = object distance; $v_0$ = image distance

---

**206. Draw a ray diagram of an astronomical telescope (refracting telescope) in normal adjustment position (image formed at infinity). State any one drawback of this type of telescope.**

A:




**Compound Microscope**

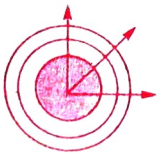
**Drawback:** Astronomical telescopes suffer from chromatic aberrations, blurred edges.

## WAVE OPTICS

**207. What is the wave front. Draw the geometrical shape of the wavefront produced by light diverging from a point source.**

A: 1) A **wave front** is the locus of all points in a medium that vibrate in the **same phase.**

2) For light diverging from a point source, the wave front is **spherical.**



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**208. If light bend around obstacles, then why can't we see around a building?**

A: 1) Light bends around obstacles due to diffraction, but diffraction is appreciable only when obstacle size is comparable to wavelength.

2) Since wavelength of light is very small, bending is negligible around buildings.

**209. Explain the statement *'light added to light can produce darkness'***

A: 1) This statement describes the phenomenon of **destructive interference**.

When two coherent light waves travel in the same direction and meet at a point out of phase, the crest of one wave falls on the trough of the other.

2) Then their amplitudes cancel out, resulting in a region of zero intensity (darkness).

---

**210. Two independent light sources cannot act as coherent sources. Then how Young performed the interference pattern on the screen?**

A: Young achieved a coherent interference pattern by using a single light source to illuminate two narrow slits. This ensured that the two secondary sources were perfectly synchronized in phase.

---

**211. Diffraction is common in sound, but not common in light waves. Why?**

A: 1) Sound waves have large wavelengths, so they diffract easily around obstacles.

2) Light waves have very small wavelengths, so diffraction is very small.

## ATOMS

**212. What happens when an electron jumps from higher energy level to lower energy level according to Bohr's model?**

A: 1) When an electron jumps from higher energy level to lower energy level, it emits a photon of electromagnetic radiation.

2) The energy of the photon is equal to the difference between the higher energy level and the lower energy level is given by $\Delta E = h\nu$.

---

**213. What is the physical meaning of negative energy of an electron?**

A : 1) The physical meaning of the negative energy of an electron in an orbit means that the electron is bound to the positive nucleus due to attractive force.

2) The energy of electron in first orbit of hydrogen is –13.6 eV. It means that the energy required to take the electron out of the hydrogen atom from first orbit is 13.6 eV.

---

**214. Name the physical quantity whose dimensions are the same as those of angular momentum?**

A: Planck's constant (h) has same dimensions as **angular momentum.**

$$E = h\nu \Rightarrow h = \frac{E}{\nu} = \frac{ML^2T^{-2}}{T^{-1}} = ML^2T^{-1}$$

## NUCLEI

**215. A fusion reaction releases energy of $1.6\times10^{-12}$J. Find the equivalent mass defect in kg.**

A: Given $E=1.6\times10^{-12}$ J. We know $c=3\times10^{8}$ m/s . We have $E=mc^2$

$$\therefore \text{Mass defect } \Delta m=\frac{E}{c^2}=\frac{1.6\times10^{-12}}{(3\times10^{8})^2}=1.78\times10^{-29}\text{kg}$$

---

**216. The radius of germanium (Ge) nuclide is measured to be twice the radius of $_4Be^9$. The number of nucleons in Ge are**

A: Nuclear radius $R=R_0A^{1/3}\Rightarrow R\propto A^{1/3}$, Given $R_{Ge}=2R_{Be}$ and also $A_{Be}=9$

$$\text{Now } \frac{^{R}G_e}{^{R}B_e}=\left[\frac{^{A}G_e}{^{A}B_e}\right]^{1/3}\Rightarrow\frac{2\cancel{R_{Be}}}{\cancel{R_{Be}}}=\left(\frac{A_{Ge}}{9}\right)^{1/3}\Rightarrow(2)^3=\left(\frac{A_{Ge}}{9}\right)\Rightarrow 8=\frac{^{A}G_e}{9}$$

$$\therefore {}^{A}G_e=8\times9=72$$

---

**217. An unstable heavy nucleus at rest breaks into two nuclei move away with velocities in the ratio of 8:27. What is the ratio of radii of the nuclei (assumed to be spherical)?**

A: From the law of conservation of momentum $m_1v_1=m_2v_2\Rightarrow\frac{m_1}{m_2}=\frac{v_2}{v_1}$.....(1)

Mass(m) is proportional to mass number (A) $\therefore\frac{m_1}{m_2}=\frac{A_1}{A_2}$.....(2)

From (1) and (2); $\frac{A_1}{A_2}=\frac{v_2}{v_1}=\frac{27}{8}$

Nuclear radius $R\propto A^{1/3}\Rightarrow\frac{R_1}{R_2}=\left[\frac{A_1}{A_2}\right]^{1/3}=\left[\frac{27}{8}\right]^{1/3}=\left[\frac{3^3}{2^3}\right]^{1/3}=\frac{3}{2}$

$\therefore$ Ratio of radii of nuclei is 3 : 2.

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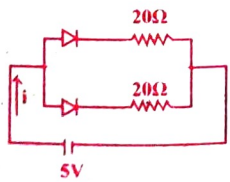
## SEMI CONDUCTORS

**218. What is the current through the battery in the circuit as shown in figure, here diodes are ideal?**




A: Both the diodes are in forward bias and both are ideal.

$R_1$ and $R_2$ are parallel. $(\because R_1=20\Omega;\ R_2=20\Omega)$

$$\therefore R_{eq}=\frac{R_1R_2}{R_1+R_2}=\frac{20\times20}{20+20}=\frac{400}{40}=10 \qquad \text{Hence, } R_{eq}=10\Omega$$

$$\therefore \text{Current } I=\frac{V}{R}=\frac{5}{10}=\frac{1}{2}=0.5A$$

# SAQ

## ELECTROSTATIC POTENTIAL

**219. Define electric field intensity and electric potential and then derive the relation between them.**

**Sol:** **Electric field intensity (E)** at a point in an electric field is defined as the force experienced by unit positive test charge ($q_0$) kept at that point. **Formula:** $E=\frac{F}{q_0}$

**Electric potential (V)** is defined as the work done per unit positive charge in bringing it from infinity to that point against the electric field.

**Relation between E and V:**

Consider two points separated by a distance 'd' in a uniform electric field of intensity E.

Let V be the potential difference between these points. The amount of work done in moving a charge q is W=Vq.....(1) The force acting on the charge q is F=Eq.

Work done W =Fd=(Eq)d.....(2) From (1) & (2), Vq=Eqd

$\therefore V=Ed \Rightarrow E=\frac{V}{d}$. This is the required relation between E and V.

---

## CURRENT ELECTRICITY

**220. Define electric resistance and write its SI unit. How does the resistance of a conductor vary if a)conductor is stretched to 4 times of its length?**

**b) temperature of the conductor is increased?**

**A:** **Electric Resistance** of a conductor is the ratio of the potential difference (V) across the ends of a conductor to the electric current (i) passing through it.

**Formula:** $R=\frac{V}{i}$. **SI unit:** ohm (Ω).

Also resistance of a conductor is given by $R=\rho\frac{l}{A}$

where ρ is resistivity of material, $l$ is length and A is area of cross section of the conductor.

**a) Conductor stretched 4 times of its Length :** When a conductor is stretched, its length increases but its area of cross section decreases so that its volume ($Al$) remains constant.

$R=\rho\frac{l}{A}=\rho\frac{l^2}{Al}$ When ρ and $Al$ are constant then $R \propto l^2$

Thus if length $l$ stretched 4 times, its resistance R becomes $4^2$ = **16 times.**

**b) When temperature of a conductor is increased,** its resistance R increases.

Resistivity of a conductor $\rho=\frac{m}{ne^2\tau}$. Here m is mass of electron, n is number of electrons per unit volume, e is charge of electron and τ is average time of collisions.

When temperature is increased, the number of collisions increases and the average time of collisions τ decreases. The term in the denominator decreases.

Thus, its resistivity and hence **resistance increases.**

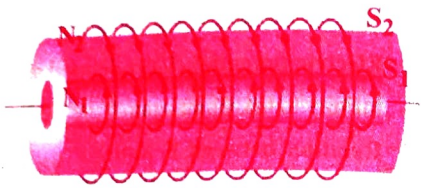
## ELECTROMAGNETIC INDUCTION

**221. Obtain an expression for the mutual inductance of two long co-axial solenoids.**

**A:** 1) **Mutual Inductance** is the phenomenon where a changing current in one coil induces an EMF in a nearby secondary coil.

Mutual induced emf $\varepsilon = -M\frac{di}{dt}$, here M is mutual inductance of the coils.

**2) Expression for mutual induction:**




Consider two co-axial solenoids $S_1$, $S_2$ each of length $l$. Let A be the common cross-section area of solenoids.

Let $N_1$ and $N_2$ be the total number of turns in the solenoids $S_1$ and $S_2$.

3) When current $I_1$ is passing through solenoid $S_1$, Magnetic field produced is $B_1 = \mu_0 \frac{N_1}{l} I_1$ .....(1)

4) Magnetic flux linked with each turn of solenoid $S_2$ is $B_1A = \left(\mu_0 \frac{N_1}{l} I_1\right)A$.

5) $\therefore$ Total magnetic flux linked with $N_2$ turns of solenoid $S_2$ is, $\phi_2 = (B_1A)N_2$

$\Rightarrow \phi_2 = \left(\mu_0 \frac{N_1}{l} I_1 A\right)N_2 = \frac{\mu_0 N_1 N_2 I_1 A}{l}$ .....(2)

6) Now total magnetic flux linked with $N_2$ turns of solenoid $S_2$ is $\phi_2 = M_{12}I_1$ .....(3) where $M_{12}$ is mutual inductance of $S_1$ with respect to $S_2$

7) From eq (2) and eq (3) we get, $M_{12}I_1 = \frac{\mu_0 N_1 N_2 I_1 A}{l} \Rightarrow M_{12} = \frac{\mu_0 N_1 N_2 A}{l}$ .....(4)

8) Let $n_1$, $n_2$ be the number of turns per unit length of the solenoids $S_1$, $S_2$

$\therefore M_{12} = \mu_0 n_1 n_2 lA$.....(5) $\left(\because n_1 = \frac{N_1}{l}, n_2 = \frac{N_2}{l}\right)$

9) Similarly, mutual inductance of solenoid $S_2$ with respect to $S_1$ is $M_{21} = \mu_0 n_1 n_2 lA$ .....(6)

Comparing eq (5) and eq (6), we get $M_{21} = M_{12}$

10) $\therefore$ Mutual inductance of two long coaxial solenoids, $M = \mu_0 n_1 n_2 lA$

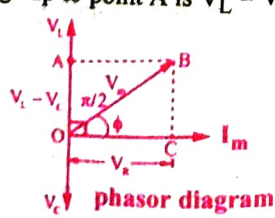
---

## ALTERNATING CURRENT

**222. Derive the expression for the impedance of a series LCR circuit by using phasor diagrams technique.**

**A:** The **Phasor diagram** for series LCR circuit is as shown in figure. Here, voltage $V_L$ is in upward direction and voltage $V_C$ is in downward direction. So, net voltage up to point A is $V_L - V_C$.

Maximum voltage is denoted by $V_m$.




From the phasor diagram, $OB = \sqrt{(OC)^2 + (CB)^2}$

$\therefore V_m = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(I_m R)^2 + (I_m X_L - I_m X_C)^2}$

$$\Rightarrow V_m = I_m\sqrt{R^2+(X_L-X_C)^2}$$

$$\frac{V_m}{I_m} = Z = \sqrt{R^2+(X_L-X_C)^2} \text{ for } X_L > X_C$$

This is the expression for impedance of series LCR circuit.

**223. Define resonance in a series LCR circuit. Derive the condition for resonance and explain what happens to current and impedance at resonance.**

**A: IN RESONANCE LCR CIRCUIT :**

An LCR circuit that allows the maximum current at a specific frequency of AC is known as a **series resonance (LCR) circuit.** Consider an LCR circuit of given frequency n.

At resonance, capacitive reactance ($X_C$)is equal to the inductive reactance ($X_L$).

Thus at resonance $X_L = X_C$

Impedance $Z = \sqrt{R^2+(X_L-X_C)^2} = \sqrt{R^2+0} = R \Rightarrow Z = R$

Again, at resonance $X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

We know $\omega = 2\pi n \Rightarrow 2\pi n = \frac{1}{\sqrt{LC}} \Rightarrow n = \frac{1}{2\pi\sqrt{LC}}$ Here n is resonant frequency.

**Current and Impedance at Resonance:** i) We know $I_0 = \frac{E_0}{R}$

$\therefore$ At resonance if R is minimum then current in the circuit becomes maximum.

ii) At resonance, $X_L = X_C \Rightarrow Z = R$.

Hence if the impedance of the circuit is minimum then resistance also becomes minimum.

## WAVE OPTICS

**224. Compare and contrast the double slit interference pattern with single slit diffraction pattern.**

**A:**

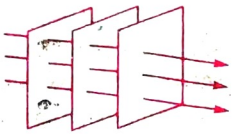
| Feature | Double slit Interference | Single slit Diffraction |
|---|---|---|
| **source** | Produced by the superposition of two separate wave fronts from two coherent sources. | Produced by the superposition of secondary wavelets originating from different parts of the same wave front. |
| **Fringe width** | All fringes are generally of equal width. | The central bright fringe is twice as wide as the secondary maxima. |
| **Intensity** | All bright fringes have the same intensity. | The intensity decreased rapidly as we move away from the central maximum. |
| **Constant** | There is usually a good constant | There is poor constant dark fringes are not perfectly dark. |

**225. Explain any two types of wave fronts. Draw the ray diagram to show the behavior of plane wave fronts as they pass through a) a thin prism b) a thin convex lens.**

A: A wave front is defined as the locus of all points is a medium that are overlapping in the same phase. There are two types of wave fronts.

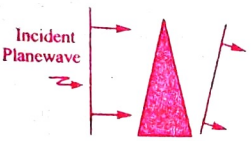
i) **Spherical wave front:** It is produced by a point source of light. As the light travels outward in all directions, the surface of constant phase is a sphere.

ii) **Plane wave front:** It is produced by a source at an infinite distance or a point source placed at the focus of a convex lens. The rays are traveled and the wave front is a plane.

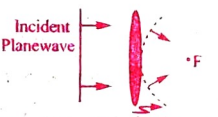


**Behaviour of plane wave front:**

a) **Through a thin prism:** When a plane wave front passes through a prism, the lower part of the wave front travels through more glass than the upper part. Since light travels slower in glass, the lower part is delayed, causting the emerging wave front to be tilted.




b) **Through a thin convex lens:** A plane wave front incident on a convex lens is refracted such that the central part of the wave front travels through the thickest part of the lens and is delayed the most. This causes the emerging wave front to become spherical and converge towards the focal point.




Spherical wavefront of radius f

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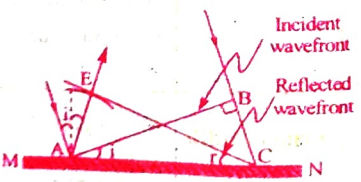
**226. State Huygen's principle. Deduce the law of reflection on the basis of Huygen's wave theory.**

A : 1) **Huygens' Principle:** Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets are called secondary wavelets and a common tangent drawn to these wavelets gives the position of wavefront at a later time.

2) **Deduction of law of reflection:** We consider a plane wave AB incident at an angle on a reflecting surface MN.




3) If 'υ' represents the speed of the wave in the medium and if 't' represents the time taken by the wave front to advance from the point B to C. Then the distance BC = υt.

4) In order to construct the reflected wave front we draw a sphere of radius υt from the point A. Let CE represent the tangent plane drawn from point C to this sphere. Then, AE = BC = υt. The triangle EAC and BAC are congruent, the angle i and r would be equal. This is the law of reflection

## DUAL NATURE OF RADIATION

**227. Explain why macroscopic objects do not show significant wave-like properties when compared to sub-atomic particles.**

A: Macroscopic objects do not show significant wave properties, because their mass is incredibly large making their associated de Broglie wavelength is very small.

De Broglie wavelength $\lambda = \frac{h}{P}$

Every moving particle has a wave nature, with a wave length λ calculated by the equation

$\lambda = \frac{h}{mv}$

Here h is planck's constant, m is mass of the particle, v is the velocity of the object.

Macroscopic objects fail to show wave properties because, the numerator (h) is small, the macroscopic mass (m) is larger than the wavelength close to zero.

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## SEMI CONDUCTOR ELECTRONICS

**228. Draw and explain the current-voltage (I -V) characteristic curves of a junction diode in forward and reverse bias.**

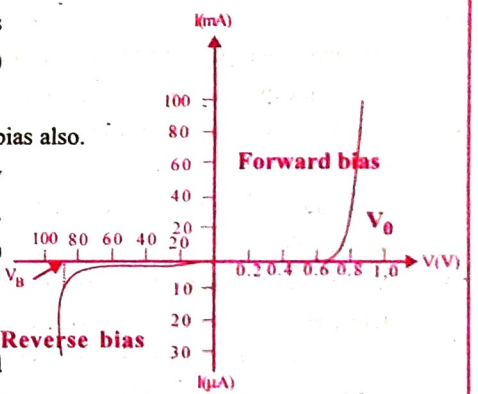
A: **Voltage - Current Characteristics of Diode:**

A diode is connected in forward bias and readings are noted with applied potential differences (PDs) and obtained currents.

The same process should be repeated in reverse bias also.

With the measured values a graph is drawn by taking **voltage** on **x-axis** and **current** on **y-axis.**

The graph obtained is called characteristic (I-V) curve of the diode.




**Forward bias - Threshold voltage ($V_0$)**

**Threshold voltage** is the voltage in a forward biased p-n junction at which the current suddenly rises to a large values.

The curve shows that the diode conducts electric current during the forward bias but can not conduct during the reverse bias. Due to this property, diode is used as a rectifier.

**Reverse bias - Breakdown voltage ($V_B$)**

**Breakdown voltage:** It is the voltage in a reverse biased p-n junction at which the current suddenly rises to a larger values.

CHAPTERS: 1,2,3,4

# Q 1 UNIT TEST-1

Time : 1 Hour

## SR. PHYSICS

Max.Marks : 25

Ans-Page Index

### SECTION-A

**I. Answer ALL the following questions.** $3 \times 1 = 3$

1. By quantization, charge q of a body is given by (where n is integer) [☞P 82(1)]
   1) q=n/2e 2)q=n(e/3) 3)q=n(2e/3) 4) q=ne
2. The SI unit of electric field is _____ [☞P 82(9)]
3. What is the SI unit of electric current? [☞P 83(11)]

### SECTION-B

**II. Answer ALL the following questions** $3 \times 2 = 6$

4. What is the importance of Gauss law in electrostatics. [☞P 62(81)]
5. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel. [☞P 63(88)]
   a) What is the ratio of charges on each capacitor?
   b) What is the ratio of potential differences across each capacitor?
6. State Ohm's Law and write its mathematical form. [☞P 65(94)]

### SECTION-C

**III. Answer ANY TWO of the following questions.** $2 \times 4 = 8$

7. State and explain Coulomb's inverse square law in electricity. [☞P 32(14)]
8. Derive an expression for the capacitance of a parallel plate capacitor. [☞P 36(23)]
9. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. [☞P 40(32)]

(or)

10. Derive an equation for the couple acting on a electric dipole in a uniform electric field. [☞P 33(16)]
11. Derive an expression for the electric potential due to a point charge. [☞P 35(20)]
12. Define electric resistance and write its SI unit. How does the resistance of a conductor vary if a)conductor is stretched to 4 times of its length? b) temperature of the conductor is increased? [☞P 99(220)]

### SECTION-D

**IV. Answer ANY ONE of the following questions.** $1 \times 8 = 8$

13. Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? [☞P 16(1)]
14. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. [☞P 17(2)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 5,6,7,8,9

# Q 2 UNIT TEST-2

Time : 1 Hour

## SR. PHYSICS

Max.Marks : 25

Ans-Page Index

### SECTION-A

**I. Answer ALL the following questions.** $3 \times 1 = 3$

1. The universal character of every material is
   1) diamagnetic 2) paramagnetic
   3) ferromagnetic 4) ferrimagnetic [☞P 84(2)]
2. The direction of induced current is determined by _____ law. [☞P 85(10)]
3. What is the time period of AC mains in India? [☞P 88(12)]

### SECTION-B

**II. Answer ALL the following questions** $3 \times 2 = 6$

4. Magnetic lines form continuous closed loops. Why? [☞P 68(110)]
5. State Faraday's law of electromagnetic induction. [☞P 69(116)]
6. What is transformer ratio? [☞P 71(125)]

### SECTION-C

**III. Answer ANY TWO of the following questions.** $2 \times 4 = 8$

7. Explain the properties of paramagnetic materials. [☞P 43(39)]
8. Explain briefly working of AC generator. [☞P 46(45)]
9. State the principle on which a transformer works. [☞P 47(47)]
   Describe the working of a transformer with necessary theory.

(or)

10. Obtain an expression for the mutual inductance of two long co-axial solenoids. [☞P 99(220)]
11. How the existence of domains, explain the behavior of ferromagnetic material? [☞P 44(40)]
12. Obtain an expression for the current through an inductor when an AC emf is applied. [☞P 48(49)]

### SECTION-D

**IV. Answer ANY ONE of the following questions.** $1 \times 8 = 8$

13. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram. [☞P 21(5)]
14. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. [☞P 23(7)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 10,11,12,13,14

# Q 3 UNIT TEST-3

Time : 1 Hour **SR. PHYSICS** Max.Marks : 25

Ans-Page Index

## SECTION-A

**I. Answer ALL the following questions.** $3 \times 1 = 3$

1. Which of the following waves cannot be polarized? [☞P 88(5)]
   1) X- rays 2) Radio waves 3) Sound waves 4) Light waves
2. The process of raising or transferring the electron from lower energy state to higher energy state is called _____ [☞P 89(8)]
3. The gap between the top of the valence band and bottom of the conduction band is called the _____ [☞P 90(8)]

## SECTION-B

**II. Answer ALL the following questions** $3 \times 2 = 6$

4. State Malus' law. Write the expression of Malus' law. [☞P 75(148)]
5. What is photoelectric effect? [☞P 76(155)]
6. Define impact parameter. [☞P 77(162)]

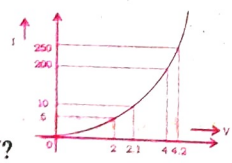
## SECTION-C

**III. Answer ANY TWO of the following questions.** $2 \times 4 = 8$

7. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. [☞P 49(51)]
8. State Bohr's postulates of hydrogen atom. [☞P 55(65)]
9. What are different types of physical processes to obtain free electrons from the metal surface. [☞P 51(55)]

(or)

10. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [☞P 49(52)]
11. What are the laws of photoelectric effect. [☞P 52(57)]
12. The I-V characteristics of a p-n junction diode in forward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V? [☞P 60(74)]




## SECTION-D

**IV. Answer ANY ONE of the following questions.** $1 \times 8 = 8$

13. Explain nuclear fission and fusion with suitable nuclear reactions. [☞P 27(10)]
14. Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance? [☞P 29(13)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

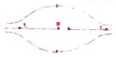
# Q4 BOARD MODEL PAPER

Time : 3 Hours **SR. PHYSICS** Max.Marks : 85

## SECTION-A

Ans-Page Index

**I. Answer ALL the following questions.** $9 \times 1 = 9$

1. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests 

   1) $E_A > E_B > E_C$ 2) $E_A = E_B = E_C$ 3) $E_A = E_C > E_B$ 4) $E_A = E_C < E_B$ [☞P 82(4)]
2. The universal character of every material is [☞P 84(2)]
   1) diamagnetic 2) paramagnetic 3) ferromagnetic 4) ferrimagnetic
3. If the flux linkage in a coil changes from 2 Wb to 8 Wb in 0.5 s, the average emf induced is: [☞P 85(5)]
   1) 10 V 2) 12 V 3) 6 V 4) 4 V
4. In an open circuit, the potential difference between the ends of a battery is equal to its _____ [☞P 83(10)]
5. What is the time period of AC mains in India? [☞P 86(12)]
6. _____ charges radiate electromagnetic waves. [☞P 87(9)]
7. Which principle is used to find the position of a wave front after some time? [☞P 88(12)]
8. The process of raising or transferring the electron from lower energy state to higher energy state is called_____ [☞P 89(8)]
9. The gap between the top of the valence band and bottom of the conduction band is called the_____ [☞P 90(8)]

## SECTION-B

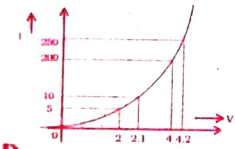
**II. Answer ALL the following questions** $14 \times 2 = 28$

10. The electric lines of force do not intersect. Why ? [☞P 62(78)]
11. In a hydrogen atom, the electron and proton are at a distance of 1Å . Find the dipole moment of the system. [☞P 64(92)]
12. Write the vector form of Ohms law. Explain the terms in it. [☞P 65(95)]
13. Write the formula for the magnetic field due to infinite straight wire and explain the terms in it. [☞P 67(107)]
14. Define magnetic susceptibility and relative permeability. [☞P 68(113)]
15. State Faraday's law of electromagnetic induction. [☞P 69(116)]
16. A Capacitor has a capacitance of $\frac{1}{\pi}\mu F$. Find its reactance for a frequency of 50 Hz. [☞P 71(128)]
17. Give two uses of infrared rays. [☞P 73(136)]
18. What is the critical angle? Give one application of total internal reflection? [☞P 74(145)]
19. State Malus' law. Write the expression of Malus' law. [☞P 75(148)]
20. Write down Einstein's photoelectric equation. Explain the terms in it? [☞P 76(158)]
21. Define impact parameter. [☞P 77(162)]
22. What are isotopes and isobars? [☞P 78(168)]
23. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? [☞P 79(176)]

## SECTION-C

**III. Answer ANY EIGHT of the following questions. 8 × 4 = 32**

24. Derive an expression for the intensity of the electric field at a point on the axial line of a dipole. |☞P 33(17)|
25. Derive an expression for the capacitance of a parallel plate capacitor. |☞P 36(23)|
26. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel. |☞P 39(28)|
    a) What is the total capacitance of the combination?
    b) Determine the charge on each capacitor if the combination is connected to a 100V supply.
27. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. |☞P 40(32)|
28. Explain the properties of paramagnetic materials. |☞P 43(39)|
29. Explain briefly working of AC generator. |☞P 46(45)|
30. State the principle on which a transformer works. Describe the working of a transformer with necessary theory. |☞P 47(47)|
31. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. |☞P 49(52)|
32. What are different types of physical processes to obtain free electrons from the metal surface. |☞P 51(55)|
33. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? |☞P 54(64)|
34. State Bohr's postulates of hydrogen atom. |☞P 55(65)|
35. The I-V characteristics of a p-n junction diode in for ward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V? |☞P 60(74)|




## SECTION-D

**IV. Answer ANY TWO of the following questions. 2 × 8 = 16**

36. (a) Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? |☞P 16(1)|
    (b)A current of 5A flows through each of two parallel long wires, the wires are 2.5 cms apart. |☞P 20(P1)|
    Calculate the force acting per unit length of each wire.
37. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. |☞P 23(7)|
38. (a) Explain nuclear fission and fusion with suitable nuclear reactions. |☞P 27(10)|
    (b) 200MeV energy is released when one nucleus of $_{92}U^{235}$ undergoes fission. Find the number of fissions per second required for producing a power of 1 megawatt. |☞P 30(P3)|

Ans-Page Index

GUESS PAPERS

# Q 5 GUESS PAPER -1

## SR. PHYSICS

Time : 3 Hours Max.Marks : 85

Ans-Page Index

## SECTION-A

**I. Answer ALL the following questions.** 9 × 1 = 9

1. The value of permittivity of free space $\varepsilon_0$ is: [☞P 82(2)]

   1) $9\times10^{9}\ N\,m^2/C^2$ 2) $8.85\times10^{-12}\ \frac{Nm^2}{C^2}$

   3) $8.85\times10^{-12}\ \frac{C^2}{Nm^2}$ 4) $9\times10^{9}\ \frac{C^2}{Nm^2}$

2. The SI unit of magnetic field intensity (H) is _____ [☞P 84(7)]
3. What is the unit of mutual inductance? [☞P 85(15)]
4. Which of the following material has the largest resistivity? [☞P 83(1)]
   1) Silicon 2) Germanium 3) Glass 4) Silver
5. According to Faraday's law, the induced emf in a coil is directly proportional to the _____ of the magnetic flux. [☞P 85(6)]
6. Which type of electromagnetic waves are used in the treatment of cancer? [☞P 87(15)]
7. The penetration of light into the region of geometrical shadow is called [☞P 88(3)]
   1) polarization 2) interference 3) diffraction 4) Refraction
8. The process of raising or transferring the electron from lower energy state to higher energy state is called _____. [☞P 89(9)]
9. In a vacuum tube, the electrons are supplied by a _____ [☞P 90(6)]

## SECTION-B

**II. Answer ALL the following questions** 14 × 2 = 28

10. The electric lines of force do not intersect. Why ? [☞P 62(80)]
11. What happens to the capacitance of a parallel plate capacitor if the distance between plates is doubled? [☞P 64(90)]
12. State Ohm's Law and write its mathematical form. [☞P 65(94)]
13. What is the importance of Oersted's experiment? [☞P 67(103)]
14. Magnetic lines of force don't intersect. Why? [☞P 68(109)]
15. State Lenz's law. [☞P 69(117)]
16. What is transformer ratio? [☞P 71(125)]
17. What is the wavelength range of X-rays? [☞P 73(141)]

18. Define power of a convex lens. What is its unit? |☞P 74(143)|
19. Compare Maxwell's Electromagnetic theory with Huygen's theory of light? |☞P 75(152)|
20. What is photoelectric effect? |☞P 76(155)|
21. Define impact parameter. |☞P 77(162)|
22. What are isotopes and isobars? |☞P 78(168)|
23. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased? |☞P 80(180)|

## SECTION-C

**III. Answer ANY EIGHT of the following questions.** $8 \times 4 = 32$

24. Derive an equation for the couple acting on an electric dipole in a uniform electric field. |☞P 33(16)||
25. Derive an expression for the electric potential due to a point charge. |☞P 35(20)|
26. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel. |☞P 37(25)|
27. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. |☞P 40(32)|
28. Explain the properties of diamagnetic materials. |☞P 43(38)|
29. Explain the Faradays law of induction and Lenz's law. |☞P 45(43)|
30. Obtain an expression for the current in a capacitor when an AC emf is applied. |☞P 48(50)|
31. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. |☞P 49(51)|
32. Explain graphically that the stopping potential is independent of intensity of incident light. |☞P 52(58)|
33. Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential. |☞P 51(56)|
34. Write a short note on debroglie's explanation of Bohr's second postulate of quantization? |☞P 56(68)|
35. Describe how a semiconductor diode is used as a half wave rectifier. |☞P 59(72)|

## SECTION-D

**IV. Answer ANY TWO of the following questions.** $2 \times 8 = 16$

36. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. |☞P 17(2)|
37. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram. |☞P 21(5)|
38. a) What is the source of stellar energy? Explain it with the help of proton-proton cycle. |☞P 27(11)|
    b) Briefly explain the formation of red giant stage of sun.

Q 6

GUESS PAPERS

# GUESS PAPER -2

Time : 3 Hours

## SR. PHYSICS

Max.Marks : 85

## SECTION-A

**I. Answer ALL the following questions.** $9 \times 1 = 9$

1. What is the SI unit of electric dipole moment? [☛P 82(14)]
2. A freely suspended magnet align in _____ direction. [☛P 84(4)]
   1) South-West 2) East-West 3) North-South 4) North-East
3. The direction of induced current is determined by _____law. [☛P 85(10)]
4. What is the SI unit of electric current? [☛P 83(11)]
5. In a series LCR circuit at resonance, the current is: [☛P 86(2)]
   1) minimum 2) maximum 3) zero 4) infinite
6. The spectrum of electromagnetic waves stretching in order of increasing wavelength from ____ rays to long ____waves. [☛P 87(6)]
7. Due to _____phenomenon of light, colours on CD are formed. [☛P 88(10)]
8. What is the SI unit of the impact parameter? [☛P 89(12)]
9. The energy band gaps of silicon and germanium are [☛P 90(2)]
   1) 0.7eV, 1.1eV 2) 1eV, 0.7eV 3) 1.1eV,0.7eV 4) 2eV, 1.5eV

## SECTION-B

**II. Answer ALL the following questions** $14 \times 2 = 28$

10. What is the importance of Gauss law in electrostatics. [☛P 62(81)]
11. Can there be electric potential at a point with zero electric intensity? Give an example. [☛P 63(84)]
12. Define temperature coefficient of resistance. [☛P 65(97)]
13. How do you convert a moving coil galvanometer into an ammeter? [☛P 67(105)]
14. Magnetic lines form continuous closed loops. Why? [☛P 68(110)]
15. Define magnetic flux. [☛P 69(118)]
16. A transformer converts 200 V AC into 2000 V AC. Calculate the number of turns in the secondary if the primary has 10 turns. [☛P 71(126)]
17. What are the applications of microwaves? [☛P 73(137)]
18. What do you understand by the focal plane? Explain with a neat ray diagram. [☛P 74(142)]
19. Two identical coherent waves, each o f intensity $I_0$, are producing an interference pattern. Write the value of resultant intensity at a point of (1) constructive interference (2) destructive interference. [☛P 75(150)]

Ans-Page Index

20. What are cathode rays? [☛P 76(153)]
21. Define distance of closest approach. [☛P 77(163)]
22. What are isotones and isomers? [☛P 78(169)]
23. What are intrinsic and extrinsic semiconductors ? [☛P 79(173)]

## SECTION-C

**III. Answer ANY EIGHT of the following questions.** $8 \times 4 = 32$

24. State and explain Coulomb's inverse square law in electricity. [☛P 32(14)]
25. Derive an expression for the capacitance of a parallel plate capacitor. [☛P 36(23)]
26. Derive an expression for the potential energy of an electric dipole placed in a uniform electric field. [☛P 36(22)]
27. State and explain Kirchhoff rules for an electric network. [☛P 41(34)]
28. Mention the properties of magnetic field lines. [☛P 43(37)]
29. Obtain an expression for the mutual inductance of two long co-axial solenoids. [☛P 100(221)]
30. Obtain an expression for the current through an inductor when an AC emf is applied. [☛P 48(49)]
31. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [☛P 49(52)]
32. What are the laws of photoelectric effect. [☛P 52(57)]
33. Write Einstein's photo-electric equation. [☛P 53(59)]
    How Millikan proved the validity of Einstein's photoelectric equation.
34. What are the limitations of Bohr's theory of hydrogen atom? [☛P 55(66)]
35. What is rectification? Explain the working of a full wave rectifier. [☛P 59(73)]

## SECTION-D

**IV. Answer ANY TWO of the following questions.** $2 \times 8 = 16$

36. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram. [☛P 18(3)]
37. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index. [☛P 22(6)]
    b) Explain the working of optical fiber with a neat diagram.
38. Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance? [☛P 29(13)]

GUESS PAPERS

# Q 7 GUESS PAPER -3

Time : 3 Hours

## SR. PHYSICS

Max.Marks : 85

### SECTION-A

**I. Answer ALL the following questions.** 9 × 1 = 9

1. SI unit of volume charge density is _____
2. Write a relation between susceptibility and relative permeability.
3. The SI unit of self inductance is:
   1) Tesla 2) Weber 3) Ampere 4) Henry
4. What are the dimensions of Resistivity?
5. The r.m.s. value of an AC of peak value $I_0$ is _____
6. Which colour of the light has the longest wavelength?
   1) Green 2) Blue 3) Red 4) Violet
7. State Huygen's principle.
8. 1eV is equivalent to
   1) $6.67\times10^{-34}$ J 2) $1.6\times10^{-27}$ J 3) $1.6\times10^{-19}$ J 4) $6.24\times10^{-22}$J
9. The pentavalent dopant donates one extra electron for conduction and hence this impurity is called _____

### SECTION-B

**II. Answer ALL the following questions** 14 × 2 = 28

10. What is meant by the statement charge is quantised ?
11. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel.
    a) What is the ratio of charges on each capacitor?
    b) What is the ratio of potential differences across each capacitor?
12. Define resistivity or specific resistance.
13. How do you convert a moving coil galvanometer into a voltmeter?
14. What is the magnetic moment associated with a solenoid?
15. What is self-induction?
16. Define power factor. On which factors does the power factor depend?
17. What is the principle of production of electromagnetic waves?
18. A small angled prism of 4° deviates a ray through 2.4°. Find the refractive index of the prism.
19. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves?

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20. Write down de-Broglie's relation and explain the terms therein. [☞P 76(159)]
21. What is the angular momentum of an electron in the second orbit of Bohr's model of hydrogen atom? [☞P 77(166)]
22. Compare the radii of the nuclei of mass numbers 27 and 64. [☞P 78(171)]
23. What is an n-type semiconductor? [☞P 79(174)]
What are the majority and minority charge carriers in it?

## SECTION-C

**III. Answer ANY EIGHT of the following questions.** $8 \times 4 = 32$

24. State Gauss's law in electrostatics and prove it. [☞P 34(19)]
25. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel. [☞P 39(28)]
a) What is the total capacitance of the combination?
b) Determine the charge on each capacitor if the combination is connected to a 100V
26. Explain the behaviour of dielectrics in an external field. [☞P 38(27)]
27. Derive equivalent emf and internal resistance of two cells in series combination. [☞P 41(33)]
28. How the existence of domains, explain the behavior of ferromagnetic material? [☞P 44(40)]
29. Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field. [☞P 45(44)]
30. State the principle on which a transformer works. [☞P 47(47)]
Describe the working of a transformer with necessary theory.
31. Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids. [☞P 50(54)]
32. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? [☞P 54(64)]
33. The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. [☞P 54(61)]
34. State Bohr's postulates of hydrogen atom. [☞P 55(65)]
35. What are n-type and p-type semiconductors? [☞P 58(71)]
Explain the process of formation of p-n junction diode.

## SECTION-D

**IV. Answer ANY TWO of the following questions.** $2 \times 8 = 16$

36. State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams [☞P 19(4)]
37 With a neat labelled diagram explain the formation of image in a simple microscope. Derive the expression for magnification when the image is formed at i) near point ii) at infinity. [☞P 25(9)]
38. a) Explain the composition of nucleus and express the atomic masses of nucleons in terms of atomic mass units. [☞P 28(12)]
b) Draw and explain the potential energy curve of pair of nucleons. Show that the density of nucleus doesn't depend on mass number A.

GUESS PAPERS

# Q 8 GUESS PAPER -4

Time : 3 Hours

## SR. PHYSICS

Max.Marks : 85

## SECTION-A

Ans-Page Index

**I. Answer ALL the following questions.** $9 \times 1 = 9$

1. The angle between the electric dipole moment $\vec{p}$ and the direction of electric field strength due to it on the equatorial line is:
   1) 0° 2) 90° 3) 180° 4) 60° [☞P 82(5)]
2. The magnetic material whose susceptibility is positive but small is _____ [☞P 84(8)]
3. What is the working principle of a transformer? [☞P 85(13)]
4. The SI unit of resistivity is _____ [☞P 83(7)]
5. What is the condition for resonance in a series LCR circuit? [☞P 86(13)]
6. Which of the following waves are also called the "Heat waves"? [☞P 87(5)]
   1) Visible light 2) X-rays 3) Infrared rays 4) Radio waves
7. What is the main condition to produce interference of light? [☞P 88(14)]
8. 1 fermi is equal to _____ meter. [☞P 8910)]
9. For a p-n junction diode, the current in reverse bias is in the order of [☞P 90(4)]
   1) Few mA 2) Few amperes 3) Few μA 4) Few kilo amperes

## SECTION-B

**II. Answer ALL the following questions** $14 \times 2 = 28$

10. Write the expression for electric intensity due to an infinitely long straight uniformly charged wire, at a radial distance r from the wire. Explain the terms in it. [☞P 91(184)]
11. Three capacitors of capacitances 1 μF, 2μF and 3 μF are connected in series. [☞P 64(89)]
    a) What is the ratio of charges? b)What is the ratio of potential differences?
12. Define mobility of charge carriers. [☞P 66(101)]
13. What is the principle of moving coil galvanometer? [☞P 67(104)]
14. Classify the following materials with regard to magnetism. [☞P 68(115)]
    Iron, Platinum, Copper, Magnesium.
15. Write the expression for self-induced emf and explain the terms therein. [☞P 69(120)]
16. Write the expression for the reactance of (i) an inductor and (ii) a capacitor. [☞P 71(127)]
17. The charging current for a capacitor is 0.6 A. [☞P 73(140)]
    What is the displacement current across its plates?
18. State the conditions under which total internal reflection takes place. [☞P 74(146)]
19. If a wave undergoes reflection and refraction, what will be the phase change in each case? [☞P 75(151)]

20. Define work function. Give an expression for it. [☞P 76(156)]
21. What is the difference between α - particle and helium atom? [☞P 77(164)]
22. What is the significance of a mass defect in a nucleus? [☞P 78(172)]
23. What is a p-type semiconductor ? What are the majority and minority charge carriers in it? [☞P 79(175)]

## SECTION-C

**III. Answer ANY EIGHT of the following questions.** $8 \times 4 = 32$

24. Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge. [☞P 32(15)]
25. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series. [☞P 37(24)]
26. Derive an expression for the energy stored in a capacitor. [☞P 38(26)]
27. Derive equivalent emf and internal resistance of two cells in parallel combination. [☞P 42(36)]
28. State Gauss' law in magnetism. Show that how this law proved the non-existence of magnetic monopoles. [☞P 44(41)]
29. Explain briefly working of AC generator. [☞P 46(45)]
30. What is power factor? Derive its expression in terms of resistance R and impedance Z. Explain its significance in AC circuits. [☞P 47(48)]
31. Explain single slit diffraction pattern with neat diagram. State conditions for maximum intensity and minimum intensity. [☞P 50(53)]
32. What are different types of physical processes to obtain free electrons from the metal surface. [☞P 51(55)]
33. What is de Broglie's wavelength? [☞P 53(60)]
34. Derive expressions for potential and kinetic energies of an electron in any orbit of a hydrogen atom according to Bohr's atomic model. [☞P 57(69)]
35. Describe energy band diagram of intrinsic semiconductors at 0 K. [☞P 58(70)]

## SECTION-D

**IV. Answer ANY TWO of the following questions.** $2 \times 8 = 16$

36. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. [☞P 17(2)]
37. Derive the lens makers formula with suitable diagram and thereby derive the lens formula. [☞P 24(8)]
38. a)What is the source of stellar energy? Explain it with the help of proton-proton cycle. b)Briefly explain the formation of red giant stage of sun. [☞P 27(11)]

GUESS PAPERS

# GUESS PAPER -5

## SR. PHYSICS

Time : 3 Hours Max.Marks : 85

## SECTION-A

**I. Answer ALL the following questions.** $9 \times 1 = 9$

1. How many number of electrons are in 1 coulomb charge?
2. Which of the following materials exhibits paramagnetic property?
   1) Iron 2) Cobalt 3) Aluminium 4) Nickel
3. The SI unit of magnetic flux is _____
4. The reciprocal of resistivity is _____
5. The power factor of a series LCR circuit at resonance is:
   1) 0 2) 0.5 3) 1 4) –1
6. What type of rays can snakes detect in addition to visible rays?
7. The phase difference between two points 'A' and 'B' on a wave front separated by distance $\lambda$ is
   1) $\pi/4$ 2) $2\pi$ 3) $\pi$ 4)$\pi/2$
8. The process of raising the atom from the normal state to the ionized state is called_____
9. Draw the circuit symbol of the p-n junction diode.

## SECTION-B

**II. Answer ALL the following questions** $14 \times 2 = 28$

10. What happens to the force between two charges if the distance between them is a) halved b) doubled
11. What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled?
12. If a wire is stretched to double its original length without the loss of mass, how will the resistivity of the wire be influenced?
13. What is the magnitude of magnetic force on a charged particle of strength q moving through a uniform magnetic field B at a velocity v which makes an angle $\theta$ with magnetic field? When does this force become maximum?
14. Define magnetisation of a material and write its formula.
15. What is the working principle of A.C. generator?
16. What is the phase difference between A.C emf and current in the following: pure resistor, pure inductor and pure capacitor.
17. If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon?

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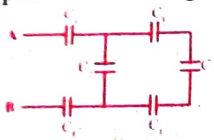
18. Two thin lens of power +4D and -2D are in contact. What is the focal length of the combination? |☞P 74(144)|
19. Diffraction is common in sound, but not common in light waves. Why? |☞P 97(211)|
20. Give examples of photosensitive substances. Why are they called so? |☞P 76(157)|
21. Name the physical quantity whose dimensions are the same as those of angular momentum? |☞P 97(214)|
22. Why are nuclear fusion reactions called thermo nuclear reactions? |☞P 78(170)|
23. What is reverse saturation current of p-n junction diode? |☞P 80(178)|

## SECTION-C

**III. Answer ANY EIGHT of the following questions.** $8 \times 4 = 32$

24. Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole. |☞P 34(18)|
25. Derive an expression for the electrostatic potential energy of a system of two point charges and find its relation with electric potential of a charge. |☞P 35(21)|
26. If $C_1 = 3$ pF and $C_2 = 2$ pF (fig.). Calculate the equivalent capacitance of the given network between points A and B. |☞P 39(29)|



27. When the resistance connected in series with the cell is halved, the current is equal to or slightly less or slightly greater than double, Why? |☞P 41(35)|
28. Define magnetisation of a sample, magnetic induction and susceptibility. How magnetic induction is related to magnetic intensity? |☞P 44(42)|
29. Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid. |☞P 46(46)|
30. State the principle on which a transformer works. Describe the working of a transformer with necessary theory. |☞P 47(47)|
31. State Huygen's principle. Deduce the law of reflection on the basis of Huygen's wave theory. |☞P 102(226)|
32. The work function of a metal is 4.3 eV. What is the threshold wavelength? |☞P 54(62)|
33. A particle has a momentum of $1.5 \times 10^{-27}$ kg m/s. What is its de-Broglie wavelength? |☞P 54(63)|
34. What are the observations of Rutherford's $\alpha$ - particle scattering experiment and how these findings supported the existence of atomic nucleus. |☞P 56(67)|
35. Describe how a semiconductor diode is used as a half wave rectifier. |☞P 59(72)|

## SECTION-D

**IV. Answer ANY TWO of the following questions.** $2 \times 8 = 16$

36. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram. |☞P 18(3)|
37. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index. |☞P 22(6)|
    b) Explain the working of optical fiber with a neat diagram.
38. Explain nuclear fission and fusion with suitable nuclear reactions. |☞P 27(10)|

# QR CODE SCANNER PAGE

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4. Moving Charges & Magnetism

5. Magnetism and Matter

6. Electromagnetic Induction

7. Alternating Current

8. Electromagnetic Waves

9. Ray Optics

10. Wave Optics

11. Dual Nature of Radiation

12. Atoms

13. Nuclei

14. Semiconductor Electronics

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**Dream Beyond Your Limits**

From a small town Rameswaram with limited means, he dreamed big, worked hard and became the People's President.

**LESSON:** Your background may be small. Your dream need not be.

**YOUR TURN:** Write one dream that you refuse to give up.

### YUVRAJ SINGH

**The Fighter Who Came Back**

After winning the World Cup, he was diagnosed with cancer. He fought bravely, completed treatment and returned to cricket.

**LESSON:** Life can knock you down. Don't stay down.

**YOUR TURN:** Don't give up because of one setback.

### AMITABH BACHCHAN

**The Phoenix**

His company failed and he faced huge losses and pressure. Instead of giving up, he started again and made a great comeback.

**LESSON:** A failure can be a chapter, not the whole story.

**YOUR TURN:** When you fail, ask – "What can I start again?"

### SINDHUTAI SAPKAL

**From Pain to Purpose**

She faced extreme hardship in life. She turned her pain into a mission and became 'Maai', mother of thousands of orphans.

**LESSON:** You can turn your pain into someone else's hope.

**YOUR TURN:** Help one person who needs you.

### MARY KOM

**I Will Try Again**

From a small village in Manipur, she faced many defeats, challenges and motherhood. But she kept coming back.

**LESSON:** A setback can stop you for a while. It need not stop you forever.

**YOUR TURN:** Try once more at something you recently gave up.

### SUDHA MURTY

**Simplicity is Strength**

She was once rejected in a job interview. She didn't lose hope and went on to become the first woman engineer in that company.

**LESSON:** Don't accept "You can't" without asking "Why not?"

**YOUR TURN:** Speak respectfully when something is unfair.

### MS DHONI

**From Railway Job to World Champion**

He worked as a ticket examiner in the railway. He kept playing cricket. Hard work changed his life.

**LESSON:** Your present situation does not decide your final destination.

**YOUR TURN:** Keep working towards your goal daily.

### SRINIVASA RAMANUJAN

**Talent Can Come From Anywhere**

He struggled in formal education, but his love for mathematics was extraordinary. His ideas amazed the world.

**LESSON:** One weakness does not cancel all your strengths.

**YOUR TURN:** Find the subject or skill in which YOU are good.

### DR. VERGHESE KURIEN

**One Idea Can Change Millions**

He was not from a farming family, but helped start the White Revolution (Amul) and changed millions of farmers' lives.

**LESSON:** Don't only think of your success, think of others too.

**YOUR TURN:** Do one thing that helps your whole group.

### RATAN TATA

**Success With Humility**

He became a top industrial leader but stayed humble and used his success to help many people.

**LESSON:** Grow in success. Stay small in ego.

**YOUR TURN:** Help someone without telling everyone.

## GOODNESS ACTIONS (NO MONEY NEEDED!)

- Plant a Sapling
- Make Someone Smile
- Read & Share Good Books
- Save Water
- Help Someone
- Keep It Clean
- Listen Patiently
- Encourage Others
- Give Without Expecting
- Respect Your Nation

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